

# Approximately inner homogeneity of pure CAR states

`MathlibAnnex.CStarAlgebra.CAR.PureStateHomogeneity`

**def**

The homogeneity property asks for exact state transport by an automorphism that is locally approximable by inner automorphisms.

## Statement

Let  $A$  be the completed CAR algebra, the norm completion of the matrix stages  $A_n = M_{2^n}(\mathbb{C})$  under  $a \mapsto a \otimes I_2$ , with the fixed coordinate reindexing. A state means a positive continuous complex-linear functional taking value 1 at the identity; a pure state is an extreme point of this convex state space. The property of pure-state homogeneity says that, for every pair of pure states  $\varphi, \psi$  on  $A$ , there is a complex star automorphism  $\alpha : A \rightarrow A$  such that  $\varphi(\alpha(a)) = \psi(a)$  for every  $a \in A$ . In addition, for every finite  $F \subset A$  and every  $\varepsilon > 0$ , there is  $u \in U(A)$  with  $\|\alpha(a) - uau^*\| < \varepsilon$  for all  $a \in F$ .

## Definition

The first requirement transports the state exactly, with the stated composition direction. The second is point-norm approximate innerness: a finite amount of algebra data can be moved almost as  $\alpha$  moves it. Both requirements concern the same  $\alpha$ . No continuous path of unitaries is part of this definition.

## Assumptions

The algebra is the fixed completed CAR algebra. The two functionals are pure states in the sense specified above. The automorphism is chosen for the state pair before the finite set and the error tolerance are given.

## Conclusion

The equality is  $\varphi \circ \alpha = \psi$ . One unitary approximates  $\alpha$  on the entire chosen finite set; that unitary may change with the finite set and tolerance.

This declaration defines a property; the separate CAR homogeneity theorem proves it. It does not assert that  $\alpha$  itself is inner or that implementing unitaries converge. The proposition about all simple separable unital  $C^*$ -algebras named Kishimoto–Ozawa–Sakai is a different, more broadly quantified source definition.

## Main citations

- Definition and its exact construction — `MathlibAnnex.CStarAlgebra.CAR.PureStateHomogeneity`
- State space and normalization — `MathlibAnnex.CStarAlgebra.stateSpace`
- Purity as extremality — `MathlibAnnex.CStarAlgebra.IsPureState`
- The proved CAR homogeneity result — `MathlibAnnex.CStarAlgebra.CAR.homogeneity`

- The separately quantified general KOS proposition —  
MathlibAnnex.CStarAlgebra.KishimotoOzawaSakaiProperty

Supporting route explanation

## Lean source signature (exact)

```
def PureStateHomogeneity : Prop :=
  ∀ (phi psi : Limit →L[ℂ] ℂ), MathlibAnnex.CStarAlgebra.IsPureState Limit phi →
    MathlibAnnex.CStarAlgebra.IsPureState Limit psi →
    ∃ alpha : Limit ≈*a[ℂ] Limit,
      (∀ a : Limit, phi (alpha a) = psi a) ∧
      ∀ (F : Finset Limit) (epsilon : ℝ), 0 < epsilon →
        ∃ v : unitary Limit, ∀ a ∈ F,
          ‖alpha a - (v : Limit) * a * star (v : Limit)‖ < epsilon
```

| In the source                                                                                                                                        | Mathematical meaning                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PureStateHomogeneity : Prop                                                                                                                          | A property of the fixed CAR algebra $A$ , rather than an automorphism or a proof that the property holds.                                                                                                                      |
| $\forall (\text{phi psi} : \text{Limit} \rightarrow L[\mathbb{C}] \mathbb{C})$                                                                       | For every pair of continuous complex-linear functionals $\varphi, \psi : A \rightarrow \mathbb{C}$ .                                                                                                                           |
| $\text{IsPureState Limit phi} \rightarrow \text{IsPureState Limit psi} \rightarrow$                                                                  | Restrict to states $\varphi, \psi$ that are positive, normalized and extreme in the real convex state space.                                                                                                                   |
| $\exists \text{alpha} : \text{Limit} \approx^*_a[\mathbb{C}] \text{Limit}$                                                                           | One bijective complex-linear star algebra map $\alpha : A \rightarrow A$ is chosen for the pair of pure states.                                                                                                                |
| $\forall a : \text{Limit}, \text{phi} (\text{alpha } a) = \text{psi } a$                                                                             | For every $a \in A$ , $\varphi(\alpha(a)) = \psi(a)$ , that is $\varphi \circ \alpha = \psi$ in this direction.                                                                                                                |
| $\forall (F : \text{Finset Limit}) (\text{epsilon} : \mathbb{R}), 0 < \text{epsilon} \rightarrow$                                                    | For every finite $F \subset A$ and every $\varepsilon > 0$ , after $\alpha$ has already been chosen.                                                                                                                           |
| $\exists v : \text{unitary Limit}, \forall a \in F, \ \text{alpha } a - (v : \text{Limit}) * a * \text{star } (v : \text{Limit})\  < \text{epsilon}$ | There is one unitary $u$ (named $v$ ) with $\ \alpha(a) - uau^*\  < \varepsilon$ for every $a \in F$ . The unitary may depend on $F, \varepsilon$ ; $\alpha$ is common to all tests. This is point-norm approximate innerness. |

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### ▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.CAR.PureStateHomogeneity

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Accepted source guide SHA-256: 6aad4a9c09c84eae33964227ced7034af6d725e23c16a2ac2a8b75f032457f21

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73